Nandan Sharma

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Professional Summary

Electronics and Communication Engineering undergraduate with hands-on experience in Python programming, software debugging, automated testing, and technical project development. Skilled in problem-solving, SQL, and REST APIs, with projects in regression test automation and AI-powered applications.

Technical Skills

- **Programming Languages:** Python, C++, SQL
- **Core Computer Science:** Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), DBMS, Operating Systems, Computer Networks
- **Software Development & Testing:** Debugging, Error Handling, Log Analysis, pytest, Regression Testing
- **Systems & Troubleshooting:** Windows, Linux Fundamentals, Hardware/Software Fundamentals
- **Backend, APIs & Databases:** Flask, FastAPI, REST APIs, MySQL, SQLite, ChromaDB
- **AI & Generative AI:** Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Ollama, LangChain, Hugging Face, OpenAI Whisper
- **Machine Learning & Data Science:** Scikit-learn, Pandas, NumPy, Data Analysis, Text Embeddings, Semantic Search, Model Evaluation
- **Developer Tools:** Git, GitHub, AWS, Docker, Command-Line Interface (CLI), VS Code, Jupyter Notebook

Experience

Infotact Solutions **Oct 2026 – Present**
Data Science & Machine Learning Intern (Data Associate L1) *Bengaluru, Karnataka*

- Perform data preprocessing, missing-value handling, and exploratory data analysis (EDA) on structured datasets using Python.
- Work on *EconoCausal: Dynamic Pricing via Double Machine Learning*, an ongoing project exploring causal effects of pricing decisions on demand.
- Apply machine learning and statistical analysis concepts to investigate data patterns and support data-driven decision-making.
- Evaluate model results and investigate potential issues to strengthen analytical and technical problem-solving skills.

Projects

Regression Guard – AI-Assisted Regression Test Generator **2026 – Present**
Python, Git, Ollama, pytest, REST APIs, Prompt Engineering, Software Testing

- Built a Python-based tool that analyzes Git commits and generates 20+ candidate regression tests for software bug fixes.
- Implemented automated test validation with `pytest` to check whether generated tests fail on buggy code and pass on the corrected version.
- Evaluated the workflow on real code changes in the open-source `humanize` library and generated validation reports to inspect test outcomes and identify issues.

Lectura AI – YouTube & Lecture Study Assistant **2026**
Python, FastAPI, Whisper, LangChain, ChromaDB, Hugging Face, Mistral AI

- Developed an audio-processing pipeline that converts lecture videos into timestamped transcripts, handling 15+ hours of audio.
- Integrated a Retrieval-Augmented Generation (RAG) pipeline using Hugging Face embeddings and ChromaDB to retrieve relevant lecture content for AI-powered Q&A.
- Tested the workflow across multiple lecture videos and integrated REST API endpoints for summaries, study questions, interactive Q&A, and content exports.

Academic Result Automation System **2025**
Python, Requests, BeautifulSoup, Flask, Automation, Error Handling

- Developed a Python automation pipeline to retrieve and structure academic result data, processing 500+ student records.
- Implemented error handling for inconsistent or unexpected result-page structures to improve the reliability of automated data extraction.
- Exported structured results to Excel and used Power BI to analyze and visualize academic performance trends.

Education

Bangalore Institute of Technology, Bengaluru, Karnataka **2023 – Present**
Bachelor of Engineering in Electronics & Communication Engineering **CGPA: 8.25/10**

Certifications & Achievements

- **INNOVAX 2K26 – Global Online Hackathon:** Secured 3rd position.

- **Tata Imagination Challenge:** Reached Round 3 and presented an original idea.
- **DSA & Problem Solving:** Solved 300+ DSA problems across LeetCode and takeUforward.
- **SQL Practice:** Solved 50 questions from LeetCode SQL 50.
- **Certifications:** Docker Essentials (IBM); Getting Started with Generative AI (IBM); Retrieval-Augmented Generation for Enhanced AI Outputs (IBM SkillsBuild); Tata GenAI Powered Data Analytics Job Simulation.

Leadership & Extracurricular Activities

- **Project Team Leader:** Leading the final-year ECG project team and coordinating project development tasks.
- **Peer Mentoring:** Helped junior students understand college coursework and prepare for examinations.
- **Sports Achievements:** Won school-level racing and cricket competitions.